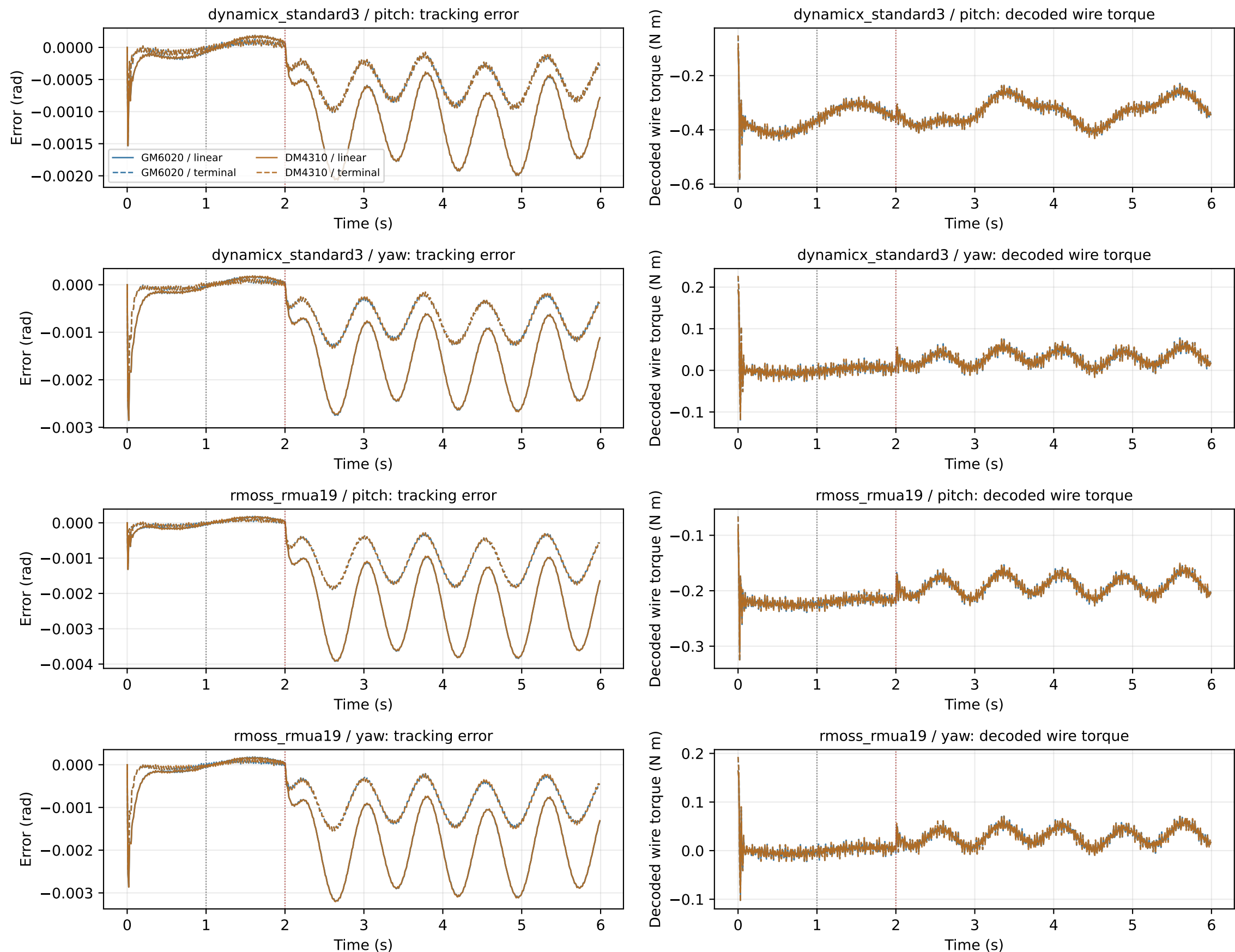


Sine: 1 kHz control, nominal J, exact model gravity feedforward



Display sampled at 100 Hz; metrics and limits checked at full simulation rates. Load step and periodic disturbance start at 2 s. Synthetic results.